National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	**Power Electronics**	**Course Code:**	**EE3032**
Program:	**BS Electrical Engineering**	**Semester:**	**Spring 2025**
Assignment Submission Date:	**9-May 2025**	**Total Marks:**	**20**
		Weight:	**2**
Section:	**BEE-6A**	**Page(s):**	**1**
Exam Type:	**Assignment- 5**	**CLO #**	**5**

Student Name:______________________________ **Roll No.**________________ **Section:**_________

Instructions:

1. Use A4-sized sheets for this assignment.
2. Avoid cutting and overwriting.
3. Secure the pages with staple, clip, or binding. Do not submit loose pages.
4. Assignment must be submitted on or before deadline.
5. **Late submission not allowed.**

Question No. 1 (CLO No. 5) **Marks: 20**

Analyze the single-phase full converter shown in figure below. The transformer secondary voltage v_s is also plotted below. Here α is the firing angle.

1. Sketch the output voltage v_o if $L = 0$ (Load is resistive)
2. Sketch the output volage v_o if $L = \infty$ (Load is inductive)
3. Determine the dc and rms value of output voltage in each case if $\alpha = 60$
4. Determine the dc and rms value of output voltage in each case if $\alpha = 30$